

Assessment of computational methods in predicting TCR–epitope binding recognition

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Supplementary Note 1: Key aspects of retraining model evaluation

Key evaluation aspects included: (1) we tested the strategies for constructing different negative data, including AS, PS and HS TCRs, and compared their performance on retrained models; (2) We assessed the cross effects in training data, including the impact of incorporating cross-reactive TCRs or not on model performance, and how sequence similarity between the training and test sets affects model performance; (3) we conducted a detailed analysis of the factors influencing model performance, including examining how the number of TCRs associated with each epitope affects model performance, analyzing the saturation point for the number of TCRs, exploring the correlation between TCR sequence similarity and prediction accuracy, and adjusting the ratio of positive to negative samples to examine its effect on model performance; (4) we separately evaluated the performance of retrained models for both seen and unseen epitope, to illustrate the generalization capabilities of prediction models.

Supplementary Note 2: Details of building test sets to evaluate the original models

To assess the effectiveness of published models for TCR-epitope binding prediction, we evaluated a total of 46 originally trained models, including 31 CDR3 β -only and 15 CDR3 β +others models (**Supplementary Table 2**). During the data preprocessing, we found that there were a large number of TCR sequences in the IEDB database that did not start with C and end with F. Through analysis, we found that the starting and ending positions of these sequences were similar to the second and penultimate positions of the TCR sequences that followed the distribution rules. In order to increase the prediction data, we added C and F at the beginning and end of these sequences that did not follow the rules¹⁸ (**Extended Data Fig. 2a, b**).

The test datasets were separately built for CDR3 β -only and CDR3 β +others models according to different feature usages. For the seen-epitope scenario of CDR3 β -only models, we initially retained the epitopes commonly used by all original models throughout training. After filtering the TCR-epitope pairings already used in model training, the test dataset (labelled as S_Data1) comprised three types of epitopes and 978 binding TCR sequences (**Fig. 2a**). For the unseen-epitope scenario of CDR3 β -only models, all unused TCR-epitope pairs during the training of all original models were retained as the test dataset. After filtering, we finally obtained 40 epitopes, corresponding to 345 TCRs in this positive test set (labelled as U_Data1) for evaluation (**Fig. 2b**). Three models (TCRGP, TCR-BERT and SETE) that trained for specific epitopes were not applicable to unseen epitopes and thus were excluded from the evaluation.

In evaluating the CDR3 β +others models of the seen-epitope scenario, after filtering TCR-epitope pairs based on the presence of CDR3 β and additional features from the training data of all collected models, the available test data was reduced to two types of epitopes, with the number of positively binding TCRs significantly decreasing to 239

(S_Data2, **Fig. 2c**). For the unseen-epitope scenario, we ultimately retained 14 epitopes corresponding to 67 bound TCRs (U_Data2, **Fig. 2d**). Models such as MixTCRpred, DeepAIR, TCRconv-large, TCRconv-small, and TCRconv-AB-large, which trained a separate model for each epitope, were excluded from the unseen-epitope evaluation.

When preparing negative datasets for testing in both seen and unseen scenarios, for CDR3 β -only models, we noticed that the negative datasets adopted in the training of these models differ in the source of TCRs (**Supplementary Table 3**), which may introduce bias in prediction. To avoid this problem, AS, PS and HS TCRs as negative datasets were constructed, respectively. In contrast, for CDR3 β +others models, only AS TCRs were used as the negative dataset in our evaluation.

Supplementary Note 3: Details of building standardized datasets to retrain the models

Beyond the assessment of prediction performance on originally trained models, we conducted a rigorous evaluation under standardized retraining conditions to impartially assess the methodological superiority of different approaches in model construction. To achieve this, we integrated the training data from all collected databases (**Methods, Supplementary Table 1**) to retrain 31 models whose training code were available, including 24 CDR3 β -only and 7 CDR3 β +others models (**Supplementary Table 2**). Among these databases, the IEDB, VDJdb, and McPAS-TCR contained the most abundant TCRs and epitopes both in terms of variety and quantity (**Fig. 3a**).

The CDR3 β -only dataset contains a total of 600 epitopes originating from nine major groups of antigens, corresponding to 98846 bound TCRs. Considering the variation in TCR sequence lengths utilized during the training across all models, we included CDR3 β sequences with length ranging from 10 to 18 (**Extended Data Fig. 4a**). To rigorously evaluate the performance of retrained models, we divided the data into two parts: (1) TCR-epitope pairs from 16 out of 19 positive TCR-epitope binding datasets, containing the majority of data (389 epitopes and corresponding 94361 TCRs) (**Extended Data Fig. 4b**), were used for five-fold cross-validation-based training and intra-database-sourced seen-epitope testing. To ensure an even representation between the training and test sets from different databases, a stratified sampling approach was adopted (**Extended Data Fig. 4c, Methods**); (2) The remaining three inter-databases formed two independent test sets, involving both seen epitopes (80 epitopes and corresponding 2941 TCRs) (**Extended Data Fig. 4d, e**) and unseen epitopes (211 epitopes and corresponding 1581 TCRs) (**Extended Data Fig. 4f, g**), providing a more stringent evaluation. Three models (TCRGP, TCR-BERT and SETE) that designed for predicting specific epitopes were not applicable to unseen epitopes and thus were excluded from the evaluation.

For CDR3 β +others dataset, a similar strategy was employed, which contained 249 epitopes originating from six major groups of antigens, corresponding to 5294 bound

TCRs with CDR3 β sequences length ranging from 10 to 18 (**Extended Data Fig. 4h**). In this dataset, 57 epitopes and 4292 bound TCRs were used for five-fold cross-validation-based training and intra-database-sourced seen-epitope testing under a stratified sampling strategy (**Extended Data Fig. 4i, j**). From inter-datasets, 18 epitopes corresponding to 313 TCRs were collected as an independent seen-epitope test set (**Extended Data Fig. 4k, l**), and 192 epitopes corresponding to 689 TCRs were gathered as independent unseen-epitope test set (**Extended Data Fig. 4m, n**). Models such as TCRconv-fullA, TCRconv-fullB, TCRconv-fullAB and TCRGP-AB, which trained a separate model for each epitope, were excluded from unseen-epitope evaluation.

Supplementary Note 4: Impact of similar sequence removal on model performance

To prevent data leakage, we implemented CD-HIT¹⁷ program to exclude TCR sequences from the test sets that shared over 95% similarity with any sequence in the training data. Comparing AUPRC scores before and after this filtering step (**Extended Data Fig. 6a-c**), we found only minor changes in AUPRC for both seen and unseen epitope scenarios using AS-based or PS-based approaches for generating negative samples. However, under the HS-based approach, removing similar sequences notably decreased AUPRC. These results would suggest that the AS-based approach, being more stringent, helps models learn accurate patterns of TCR-epitope binding and mitigates evaluation bias from sequence similarity. In contrast, the HS-based approach appears more sensitive to TCR sequence similarity, possibly due to confounding factors such as batch effects, which can cause overfitting to irrelevant features rather than genuine binding interactions. The PS-based approach, which includes only clonally expanded TCRs from tumor tissues, is more likely enriched in tumor antigen associated TCRs. This may reduce false negatives and allow models to learn relatively robust features of TCR-epitope recognition, thereby minimizing the impact of noise arising from non-antigen-specific TCR characteristics, when compared with HS-based results.

Supplementary Note 5: Low prevalence effect of true TCR-epitope pairs on retrained models

Given the vast diversity of the TCR repertoire in humans and the real-world rarity of true specific TCR-epitope bindings (with a prevalence rate as low as 0.1%), evaluating models under conditions that mimic this low prevalence is crucial. Assessing the positive predictive value (precision) of these models holds significant practical importance, as it reflects their ability to accurately identify genuine TCR-epitope interactions amidst numerous false positives.

To systematically evaluate model performance across different prevalence scenarios, we down-sampled test and independent test sets 20 times at varying prevalence rates (0.1%, 1%, 10% and 50%) while maintaining a 1:1 positive-to-negative ratio in training

data to balance sensitivity and specificity (**Extended Data Fig. 8a-c**). This method simulates real-world conditions where true TCR-epitope pairs are exceptionally rare.

In the seen-epitope scenario, when tested with an equal ratio of positive to negative samples (50% prevalence), several models—such as epiTCR, TEPCAM, TCRGP, TCR-BERT, and TEIM—achieved high precision scores of around 0.7, along with balanced recall and specificity (**Extended Data Fig. 8a, b**). However, almost all models showed a sharp decline in precision as prevalence decreased. At the extreme prevalence of 0.1%, precision scores plummeted to 0.001-0.003, indicating significant struggles in predicting rare true TCR-epitope interactions. Models like TEINet, DeepTCR, ERGO-Istm, TITAN, and MCMC exhibited imbalanced recall-specificity trade-offs, either achieving high recall with low specificity or vice versa, and showed a marked drop in precision with decreasing prevalence.

In the unseen-epitope scenario, all models achieved low precision scores (<0.52), likely due to limited generalization capacity (**Extended Data Fig. 8c**). The majority also showed a significant decrease in precision as prevalence decreased, consistent with the results from the seen-epitope scenario. These findings highlight that, despite being trained on balanced data, existing models struggle to maintain high precision under skewed class distributions typical of real-world applications.

Supplementary Note 6: Evaluation of antigen-specific TCR identification methods on retrained models

Currently, positive data for TCR-epitope binding prediction are derived from various antigen-specific TCR identification methods. Due to differences in experimental principles and detection techniques, these methods may yield varying false positive rates, affecting both data reliability and model performance. To evaluate the quality of these data, we analyzed the annotation of TCR-epitope pairs and classified them into two main categories based on the experimental approaches used: multimer-based assays and in vitro stimulation-related experiments. In the multimer dataset, cross-reactive TCRs account for 2.6% of the entries, while in the in vitro stimulation dataset, the proportion is at 2.7%, indicating a comparable level of cross-reactivity between the two methods. However, these estimates are derived from the subset of data with explicit experimental method annotations, which constitutes 15% of the total TCR-epitope binding pairs used in this study.

We evaluated the reliability of these methods using two complementary strategies (**Methods**). First, we applied 10 top-performing models (from **Fig. 3d**), under the assumption that consensus predictions across multiple high-performing models may indicate higher confidence in true TCR-epitope interactions, to estimate the false positive rates (FPRs) of samples derived from each experimental method. Based on this approach, data generated by in vitro stimulation exhibited a lower FPR compared to those from multimer-based assays (**Supplementary Fig. 2a**). Second, we trained models separately on positive data derived from each method and tested them on a shared, high-confidence external dataset consisting of unseen epitopes. After

controlling for data size and randomness (10 repeated runs), models trained on dataset from these two methods achieved comparable predictive performance (**Supplementary Fig. 2b**).

These results suggest that, within the scope of this evaluation, TCR-epitope annotation derived from in vitro stimulation may exhibit relatively lower false positive rates, though the differences in downstream model performance are less pronounced. It is important to note that our findings are limited by the small proportion of TCR-epitope pairs with explicitly annotated experimental sources and the uncertainties in model performance—particularly in the context of unseen-epitope prediction where the results approach randomness under the second strategy. These limitations highlight the challenges of investigating false positives in experimental datasets solely through model-based evaluations. To rigorously assess the reliability of different experimental methods, additional high-confidence experimental validations are required.

Supplementary Note 7: Details of correlation analysis between model performance and sequence dissimilarity

We also computed the correlation between predictive performance and sequence dissimilarity, measured by Levenshtein distance, for these top 10 models. Almost all models exhibited a negative correlation, suggesting that higher sequence similarity among TCRs improves performance (**Extended Data Fig. 9d**). To quantitatively compare the magnitude of this effect across different models, we applied Fisher's r-to-z transformation and performed multiple testing correction on p-values. Our results indicate no significant differences in the strength of this correlation among the top 10 models (**Extended Data Fig. 9e**). However, certain models, like PiTE, showed relatively more pronounced deviations, though not statistically significant. Overall, these analyses would suggest that most top-ranked models perform similarly when handling heterogeneous TCR sequences, although some may be less sensitive to sequence variability.

Supplementary Note 8: Detailed description of data sources

For constructing the positive dataset, the following data sources were utilized:

(1) *VDJdb Database*³⁰: A widely recognized resource for TCR-epitope binding research, containing TCR sequences with varying annotation levels. Some entries include full-length α and β chains, V, D, J gene usage, as well as detailed description of experimental methods. Each TCR is assigned a confidence score (0–3) based on source reliability and annotation completeness. The June 2024 release was utilized in this study.

(2) *McPAS-TCR Database*³¹: A curated collection of TCR sequences associated with various pathologies and antigens, encompassing a wide range of human and murine diseases, such as infections, cancer, and autoimmune conditions. A subset of entries

provide complete TCR information (α and β chains, V, D, J genes) and experimental validation.

(3) *IEDB Database*³²: A comprehensive repository of experimentally validated antibody and T-cell epitope data across multiple species, covering contexts such as infectious disease, allergy, autoimmunity, and transplantation. Some records contain full TCR α and β chains with V, D, and J gene annotations. The database is updated bi-monthly.

(4) *PIRD Database (TBAdb)*³³: A specialized dataset within the Pan Immune Repertoire Database, focused on storing TCR and BCR sequences associated with specific diseases and antigens. A portion of the TCR entries include α and β chains, V, D, and J genes, and information of experimental methods.

(5) *dbPepNeo2.0 Database*³⁴: A tumor neoantigen database that categorizes entries by confidence level (high, medium, low), determined by immunogenicity testing or mass spectrometry validation. TCR-related information includes CDR3 β sequences and the corresponding V β , D β , and J β genes.

(6) *ImmuneCODE Database (MIRA)*³⁵: A large-scale dataset containing TCR repertoires from over 1,400 SARS-CoV-2-infected or -exposed individuals. Antigen-specific TCRs were identified using the MIRA assay. The TCR sequence data include CDR3 β regions along with the corresponding V β and J β gene information.

(7) *Glanville et al. (Glanville-2017)*³⁶: A study that profiled antigen-specific TCRs from 28 donors infected with Epstein-Barr virus and cytomegalovirus, with experimental techniques to identify corresponding peptides. A portion of the samples contains α and β chain sequences.

(8) *Tsuruta et al. (Tsuruta-2018)*³⁷: An investigation into CD4⁺ TCR recognition of long peptides derived from bladder cancer-associated CTAs, supported by in vitro experimental evidence. The dataset contains α and β chain sequences along with V and J gene annotations for a subset of TCRs.

(9) *Luo et al. (Luo-2018)*³⁸: A study exploring how TCR clonotypes recognize orexin and influenza peptides, with autoimmunity implications. Tetramer staining was used to validate peptide binding. The dataset includes α and β chains with V and J gene annotations.

(10) *Zhang et al. (TetTCR-2018)*³⁹: A dataset generated using TetTCR-seq, profiling CD8⁺ TCR responses to viral and diabetes-associated antigens. It includes α and β chain sequences along with V and J gene usage information.

(11) *Huth et al. (Huth-2019)*⁴⁰: An analysis of CMV-specific CD8⁺ T cells based on TCR β sequence characteristics. Peptide-MHC tetramer enrichment was used to identify CDR3 β sequences, along with their V β and J β gene usage.

(12) *Ma et al. (TetTCRHD-2021)*⁴¹: A high-dimensional TCR profiling study using TetTCR-SeqHD to identify CD8⁺ TCRs reactive to diverse antigens. The dataset includes α and β chains with V and J gene annotations.

(13) *Francis et al. (Francis-2022)*⁴²: A study examining CD8⁺ T cell responses to SARS-CoV-2 at different disease stages using peptide-based assays. A total of 648 antigen-specific TCRs restricted by four HLA alleles were identified, with α and β chain sequences and V and J gene annotations provided.

(14) *Lu et al. (pMTnet-2021)*⁴³: This study used the experimentally verified TCR-epitope paired data collected from multiple articles. The dataset only contained the CDR3 β region of the TCR sequence.

(15) *Ishigaki et al. (Ishigaki-2022)*⁴⁴: A study investigating the impact of HLA alleles on antigen recognition via TCR hypervariable regions. It includes validated TCR-epitope pairs with CDR3 β , V β , and J β gene annotations.

(16) *Minervina et al. (Minervina-2022)*⁴⁵: An experimental study that identified CD8⁺ T cell clones reactive to SARS-CoV-2 peptides restricted by various HLA alleles. The dataset includes α and β chain sequences along with V and J gene usage information.

(17) *Mudd et al. (Mudd-2022)*⁴⁶: A dataset derived from analysis of T follicular helper (Tfh) cells following SARS-CoV-2 mRNA vaccination. It provides antigen-specific α and β chain sequences with V and J gene annotations.

(18) *Feng et al. (PISTE-2024)*⁴⁷: This study utilized experimentally validated TCR-epitope pairings compiled from multiple publications (Testset II). The dataset contains only the CDR3 β region of the TCR sequences.

(19) *Nielsen et al. (IMMREP23)*¹⁴: A benchmark dataset created for evaluating TCR-pMHC specificity prediction models (Test set). It includes validated TCR-epitope pairs with α and β chain sequences and the corresponding V and J genes.

For constructing the negative dataset, the following external TCR sources were utilized:

(20) *TCRdb2.0 Database*⁴⁸: A large-scale database comprising over 691 million TCR β sequences from 19,701 samples across 269 projects. TCRdb provides detailed information on TCR β chains, and V-D-J related genes, organized by tumor type and cell type. The patient-sourced TCRs were obtained from this database.

(21) *Dean et al. (Dean-2015)*⁴⁹: A study that used multiplex PCR to amplify TCR β sequences from peripheral blood samples of 587 healthy individuals. To minimize the inclusion of CMV-specific TCRs in the negative set—which could introduce false negatives—CMV-positive samples were excluded, and only CMV-negative individuals from the healthy cohort were selected to construct the healthy-sourced negative dataset in our study.

Supplementary Note 9: Description of evaluated models

The brief summary for each model included in our benchmark study is as follows:

- (1) *ATM-TCR*⁵⁰: A deep learning model utilizing a multi-head attention mechanism. Its strength lies in predicting rare epitopes and revealing that TCRs binding to the same epitope share similar attention weight distributions, making it ideal for comparing the characteristics of different TCRs binding to the same epitope. CDR3 β and epitope sequences were used as inputs to the model.
- (2) *AttnTAP-mc*⁵¹: A deep learning model integrating an attention mechanism. It uses a BiLSTM (Attn-BiLSTM) to extract CDR3 β features and an MLP (Attn-MLP) to extract epitope features. These features are then concatenated and fed into an MLP network to predict the likelihood of TCR-epitope binding. This model was trained on the McPAS-TCR database.
- (3) *AttnTAP-vdj*⁵¹: Follows the same framework and training process as *AttnTAP-mc*, but was trained on data from the VDJdb database.
- (4) *DeepAIR*²¹: A deep learning framework that first extracts features with multichannel encoders, then fuses feature representation via a gating-based attention mechanism. At last, task-specific layers are designed to map the generated representation to outputs. This approach enables integrating sequence (TCR CDR3 α , CDR3 β , V α , V β , J α and J β) and predicted structure feature (TCR CDR3 α and CDR3 β) obtained by AlphaFold to perform immune receptor binding prediction. This tool provided a separate model for each epitope.
- (5) *DeepTCR*⁵²: A deep learning framework that combines unsupervised and supervised models to analyze TCR sequencing data for both descriptive characterization and prediction of TCR specificity, using CDR3 β and epitope sequences as input.
- (6) *DeepTCR-ABVJ*⁵²: An extended version of *DeepTCR* that incorporates additional input features, including TCR CDR3 α , CDR3 β and the corresponding V and J gene segments (V α , V β , J α , and J β), to enhance predictive performance.
- (7) *DLpTCR-FULL*⁵³: An ensemble deep learning framework focusing on predicting unseen epitopes. It compares different encoding methods, including one-hot encoding, encoding based on 500 physicochemical properties of amino acids, and a hybrid encoding. The model uses 20 principal components (PCs) derived from PCA of the physical and chemical feature set and feeds these into the LeNet-5 classifier for the final classification task. This model accepts either CDR3 β or CDR3 α sequences, along with epitope sequences, as input features.
- (8) *DLpTCR-CNN*⁵³: Similar to *DLpTCR-FULL* but uses 18 PCs for feature encoding and employs an FCN (Fully Connected Network) as the classifier.
- (9) *DLpTCR-RESNET*⁵³: Similar to *DLpTCR-FULL* but uses 10 PCs for feature encoding and employs a RESNET (Fully Residual Network) as the classifier.

- (10) *epiTCR*⁵⁴: A machine learning model built using the random forest algorithm. It encodes both CDR3 β and epitope features using the Blosum62 matrix, focusing on imbalanced classification with emphasis on sensitivity and specificity.
- (11) *epiTCR-BH*⁵⁴: Follows the same training method as *epiTCR*, with the addition of MHC class information as input features.
- (12) *ERGO-AE-vdj*⁵⁵: A deep learning model using an autoencoder accepts the CDR3 β and epitope sequences as feature inputs. This model utilizes the VDJdb dataset and employs an AutoEncoder (AE) framework during training.
- (13) *ERGO-AE-mc*⁵⁵: Similar to *ERGO-AE-vdj* but uses the McPAS-TCR dataset for model training.
- (14) *ERGO-lstm-vdj*⁵⁵: Similar to *ERGO-AE-vdj* but uses Long Short-Term Memory (LSTM) for feature encoding instead of an AE.
- (15) *ERGO-lstm-mc*⁵⁵: Similar to *ERGO-lstm-vdj* but uses the McPAS-TCR dataset for model training.
- (16) *ERGO-II-vdj*⁵⁶: An updated version of ERGO that incorporates additional features beyond the CDR3 β sequence, including the CDR3 α sequence, V α , V β , J α , J β , MHC type, and T-cell subtype (CD4/CD8). TCRs are encoded using an autoencoder, while peptides are encoded with an LSTM network. This model uses VDJdb dataset for training.
- (17) *ERGO-II-mc*⁵⁶: Similar to *ERGO-II-vdj* but uses the McPAS-TCR dataset for model training.
- (18) *ImRex*¹²: A deep learning model using CNNs. It constructs a 2D interaction map of the physicochemical properties of individual amino acids at different positions in the TCR CDR3 β and epitope sequences, akin to image classification tasks. Specifically designed for predicting both known and unseen epitopes.
- (19) *iTCep*⁵⁷: A deep learning framework that takes CDR3 β and epitope sequences as inputs and improves the performance of TCR-peptide binding prediction by introducing a feature-level fusion strategy.
- (20) *MCMC*⁵⁸: A hybrid convolutional neural network model that applies deep metric learning and multimodal learning in predicting TCR-epitope binding interactions by employing TCR CDR3 β and epitope sequences as input features.
- (21) *MixTCRpred*²⁴: A transformer-based classification model that requires CDR3 α , CDR3 β , V α , V β , J α , and J β as inputs to predict the binding score targeting a given pMHC. CDR1 α , CDR1 β , CDR2 α , and CDR2 β are optional features. This tool provided a separate model for each epitope.
- (22) *NetTCR*²⁵: A machine learning model based on a Convolutional Neural Network (CNN). It encodes features using the Blosum50 matrix and extracts features via a 1D

CNN for classification. This model used TCR CDR3 β and epitope sequences as inputs.

(23) *NetTCR-AB*²⁵: An extension of *NetTCR* that follows the same training framework but incorporates both TCR CDR3 α and CDR3 β sequences along with the epitopes as input features.

(24) *PanPep*⁵⁹: Incorporate two learning modules to predict TCR-peptide binding. The meta-learning module develops peptide-specific learners by capturing key features of CDR3 β -peptide interactions, while the disentanglement distillation module builds connections between peptide embeddings and the peptide-specific learners. This framework supports predictions across three distinct settings: few-shot learning, zero-shot learning, and majority learning, ensuring broad applicability and adaptability to various data scenarios.

(25) *PISTE-random*⁴⁷: PISTE model is short for Physics-Inspired Sliding Transformer, this model enhances TCR-antigen binding prediction by refining the conventional attention mechanism with key physical and biological priors. It consists of a sequence encoder that extracts features from TCR CDR3 β , epitope, and HLA sequences, a sliding attention module to capture interactions between features, and an alignment-based pooling module that generates input representations for predicting binding status. This model used random sampling as negative sample generation approach.

(26) *PISTE-unipep*⁴⁷: Built on the same framework as *PISTE-random*, this version takes a different negative sampling approach, named unified epitope negative sampling, which selects epitopes based on their frequency distribution within the positive dataset.

(27) *PISTE-reftc*⁴⁷: Built on the same framework as *PISTE-random*, this version takes a different negative sampling approach, named reference TCR negative sampling, which pairs each epitope with TCRs randomly sampled from a healthy donor-derived reference dataset.

(28) *PiTE-base-tcrSplit*⁶⁰: This model uses a pretrained language model to generate amino acid sequence embeddings, followed by sequence encoders and linear layers for classification. It takes TCR CDR3 β and epitope sequences as input features and ensures that no TCRs in the test set appear in the training set.

(29) *PiTE-base-epiSplit*⁶⁰: Built on the same framework as *PiTE-base-tcrSplit*, this version ensures that no epitopes in the test set are present in the training data. In our retraining, we employed TCR-BERT for feature extraction and average pooling as the sequence encoder, as the original pretrained amino acid embedding model was not publicly available.

(30) *pMTnet*⁴³: A transfer learning-based model for predicting TCR-neoantigen/antigen pairing, reached accurate results using only TCR CDR3 β , MHC allele types, and peptide sequences.

(31) *pMTnet_omni*⁶¹: An extension of *pMTnet*, incorporating additional features such as CDR3 α , V α , V β , J α , and J β .

(32) *SETE*⁶²: A machine learning model leveraging decision tree ensembles. It uses a K-mer method for feature extraction to avoid bias caused by varying TCR sequence lengths while preserving internal sequence information. SETE trains separate gradient boosting models for each epitope using TCR CDR3 β and epitope sequences, which limits its ability to generalize to unseen epitopes.

(33) *TCR-BERT*⁶³: A deep learning model built on the Bidirectional Encoder Representations from Transformers (BERT) architecture. It uses self-supervised learning to generate TCR CDR3 β sequence embeddings applicable to various tasks, including predicting TCR-epitope interactions and enabling clustering analysis of TCR sequences. This tool trains a separate model for each epitope.

(34) *TCRconv-small*⁶⁴: A deep learning model that employs convolutional neural networks (CNNs) to learn sequence-based features and motifs essential for binding specificity. It uses TCR CDR3 β and full-length TCR β sequences, along with epitope sequences, as inputs, and is trained on a small, high-quality dataset.

(35) *TCRconv-large*⁶⁴: Based on the same framework as *TCRconv-small*, this model is trained on a larger dataset that includes TCRs across all confidence levels.

(36) *TCRconv-AB-large*⁶⁴: Also built on the *TCRconv-small* framework, this model incorporates TCR CDR3 α , full-length TCR α sequences, TCR CDR3 β , and full-length TCR β sequences as inputs, and is trained on a large, confidence-inclusive dataset.

(37) *TCRfinder*⁶⁵: A deep learning-based pipeline designed for TCR and neoantigen screening. This method leverages two language models and a transformer architecture to generate high-quality representations and embeddings for CDR3 β and peptide sequences. These embeddings are then processed through a multi-layer perceptron to accurately score TCR-peptide pairings.

(38) *TCR-H*⁶⁶: A support vector machine model for predicting TCR-epitope specificity. This model takes CDR3 β sequences and a set of physicochemical features generated based on peptide sequences as inputs and only the experimentally validated negative samples in model training.

(39) *TCRGP*⁶⁷: A machine learning model based on Gaussian Processes (GP) that encodes TCR features using the Blosom62 matrix and employs a Gaussian classifier for prediction. The framework trains an individual model for each epitope, with CDR3 β as input.

(40) *TCRGP-AB*⁶⁷: Follows the same training method as *TCRGP*, but with both CDR3 α and CDR3 β as inputs.

(41) *TEIM*⁶⁸: Stands for TCR-Epitope Interaction Modeling, a deep learning-based model consisting of two submodules: TEIM-Res for modeling residue-level interaction conformations between TCRs and epitopes, and TEIM-seq for identifying whether

TCRs and epitopes bind. By taking TCR CDR3 β and epitope sequences as input features, this model enables the identification of critical binding sites between TCRs and epitopes.

(42) *TEINet-small*⁶⁹: A deep learning framework employing transfer learning. It uses two pretrained encoders to convert CDR3 β and epitope sequences into numerical vectors, concatenates the features, and feeds them into a three-layer neural network. The model emphasizes the impact of negative sample splitting strategies on performance. This model was trained on a dataset (~40,000 pairs) from VDJdb and McPAS-TCR databases.

(43) *TEINet-large*⁶⁹: Shares the same framework and training process as *TEINet-small*, but was trained on a larger dataset (~100,000 pairs) from VDJdb, McPAS-TCR, and IEDB databases.

(44) *TEPCAM*²³: Combines convolutional neural networks (CNNs) and attention mechanisms for TCR-epitope binding prediction. This model takes CDR3 β and epitope sequences as inputs.

(45) *TITAN*⁷⁰: Leverage a bimodal neural network to encode CDR3 β and epitope sequences and adopts transfer learning to boost prediction performance.

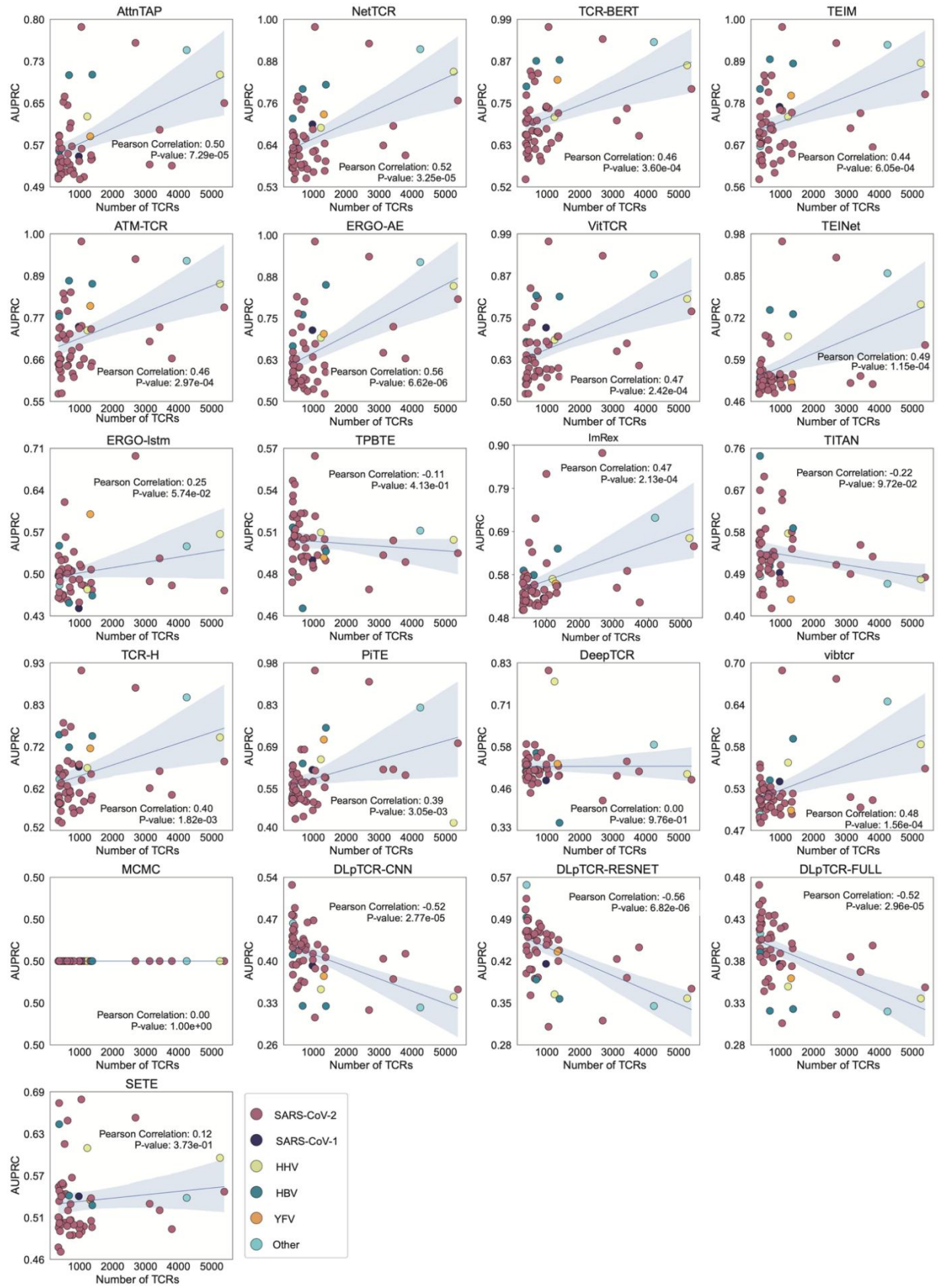
(46) *TPBTE-vdj*⁷¹: A deep learning model based on a convolutional transformer architecture. It incorporates a convolutional transformer encoder and a cross-attention fusion module to extract both local and global features of TCR CDR3 β and epitope sequences. Trained using data from the VDJdb database.

(47) *TPBTE-mc*⁷¹: Follows the same framework and training process as *TPBTE-vdj* but is trained on data from the McPAS-TCR database.

(48) *vibtc*⁷²: Applied Attentive Variational Information Bottleneck (AVIB) model to TCR-peptide interaction prediction, where Attention of Experts (AoE) is proposed to estimate the posterior distribution of latent encodings. This model takes CDR3 β and epitope sequences as inputs.

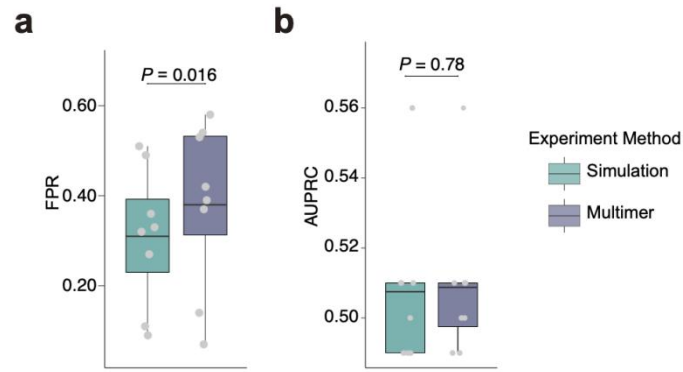
(49) *vibtc-AB*⁷²: Based on the same AVIB framework as *vibtc*, but utilizes both TCR CDR3 α and CDR3 β sequences as well as epitopes as input features.

(50) *VitTCR*⁷³: A vision transformer-based architecture adapted for TCR-antigen prediction, leveraging attention mechanisms for CDR3 β and peptide sequence feature embedding.



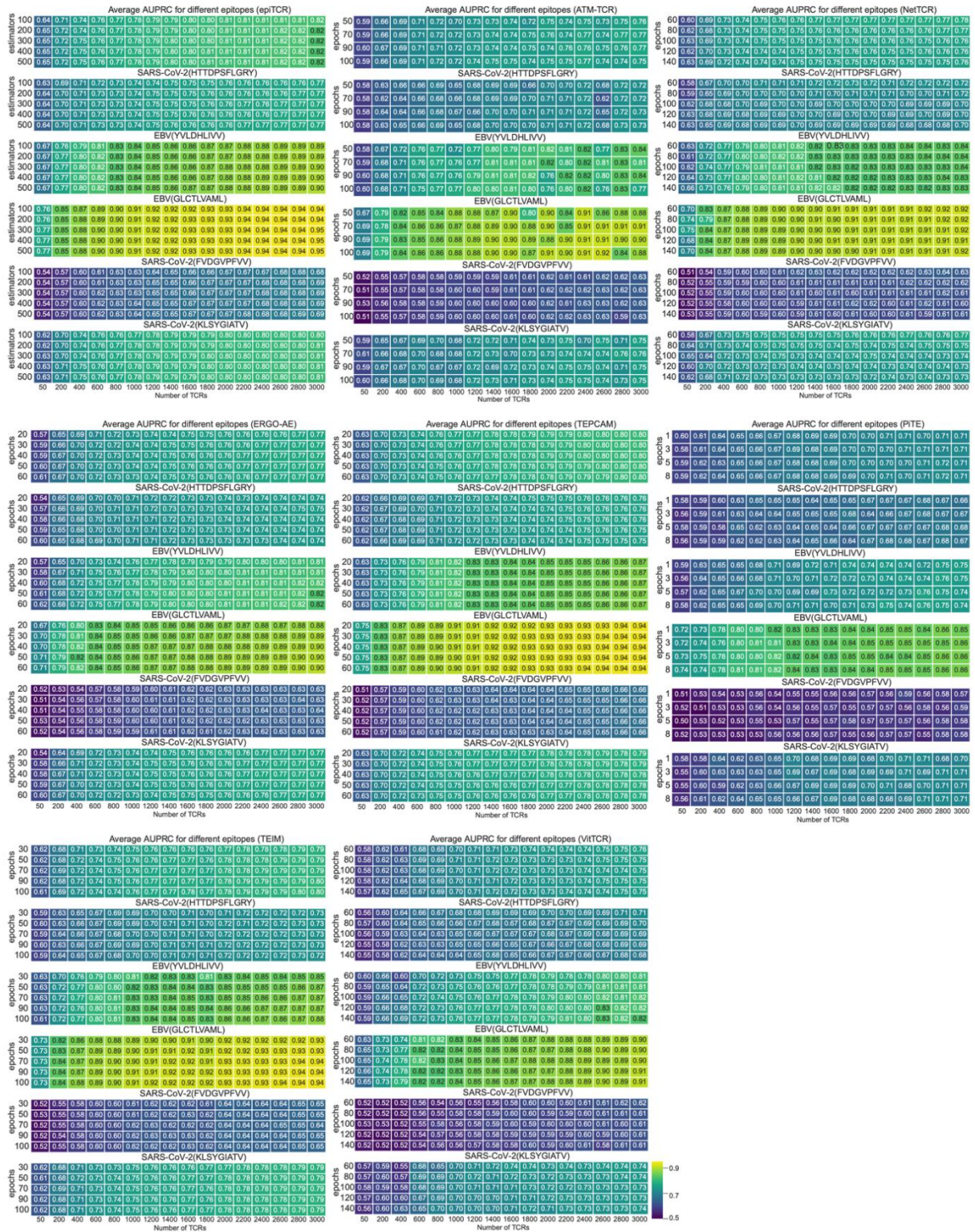
Supplementary Fig. 1: Correlation analysis between epitope-associated TCR counts and AUPRC for models: AttnTAP, NetTCR, TCR-BERT, TEIM, ATM-TCR,

ERGO-AE, VitTCR, TEINet, ERGO-Istm, TPBTE, ImRex, TITAN, TCR-H, PiTE, DeepTCR, vibtcr, MCMC, DLpTCR-CNN, DLpTCR-RESNET, DLp-TCR-FULL, SETE. Correlation between epitope-associated TCR counts and AUPRC for top-3 models from **Fig. 5a** of >300 group, dots represent epitopes, colored by antigen group. *P*-values of Pearson correlations were from two-sided *t*-test ($n=53$).



Supplementary Fig. 2: Evaluation on different antigen-specific TCR identification methods with retrained models.

a, False positive rates of TCR-epitope pairings identified with different methods predicted by 8 high-performing CDR3 β -only models retrained using AS negatives, where random sampling was adopted in negative data generation as the data could not provide the features required by our refined reshuffling approach. **b**, Performance of CDR3 β -only models retrained with the data identified by two methods: multimer-based assay and in vitro simulation, where the training sample size matched the data size of in vitro simulation. Box plots (**a**, **b**) display mean (center line), the first and third quartiles (box), minimum and maximum values within 1.5 $\times$ interquartile range (whiskers). *P*-values are from two-sided Wilcoxon signed-rank tests ($n=8$).



Supplementary Fig. 3: Results of parameter tuning when testing the effects of TCR counts on model performance: epiTCR, TEPCAM, ViTTCR, TEIM, ERGO-AE, PiTE, NetTCR, and ATM-TCR. There is a shared x-axis denoting the number of TCRs and 5 heatmaps presenting the AUPRC values of models built with different epitopes, and the top heatmap shows the average AUPRC values of models obtained by 5 epitopes.